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ME469: Review of Partial Differential Equations

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Equation Review

Real World

- The intended application

Conceptual
Model

- Physical laws, hypothesis and models (e.g. a set of PDEs) and initial and boundary conditions

Computer
Model

- A set of computational algorithms that allow to build a numerical, or approximated solution to the conceptual model



Partial Differential Equation: Transient Evolution of a Scalar with Advection, Diffusion, and Source

General Transport Equation using Einstein notation, i.e., repeated index, for scalar ϕ

- Density, ρ [kg/m³]
 - Velocity, u_j [m/s]
 - Mass diffusivity, D [m²/s²]
 - Dynamic viscosity, μ [kg/(m-s)]
 - Schmidt number, Sc
- $$\frac{\partial \rho \phi}{\partial t} + \frac{\partial \rho u_j \phi}{\partial x_j} - \frac{\partial}{\partial x_j} \left(\rho D \frac{\partial \phi}{\partial x_j} \right) = S^\phi \quad \rho D = \frac{\mu}{Sc}$$

$$\frac{\partial \rho \phi}{\partial t} + \frac{\partial \rho u_j \phi}{\partial x_j} - \frac{\partial}{\partial x_j} \left(\frac{\mu}{Sc} \frac{\partial \phi}{\partial x_j} \right) = S^\phi$$

Time term, or "accumulation"

advection

diffusion

source/sink term

Repeated Index Example

$$\frac{\partial \rho u_j \phi}{\partial x_j} = \frac{\partial \rho u_x \phi}{\partial x} + \frac{\partial \rho u_y \phi}{\partial y} + \frac{\partial \rho u_z \phi}{\partial z}$$



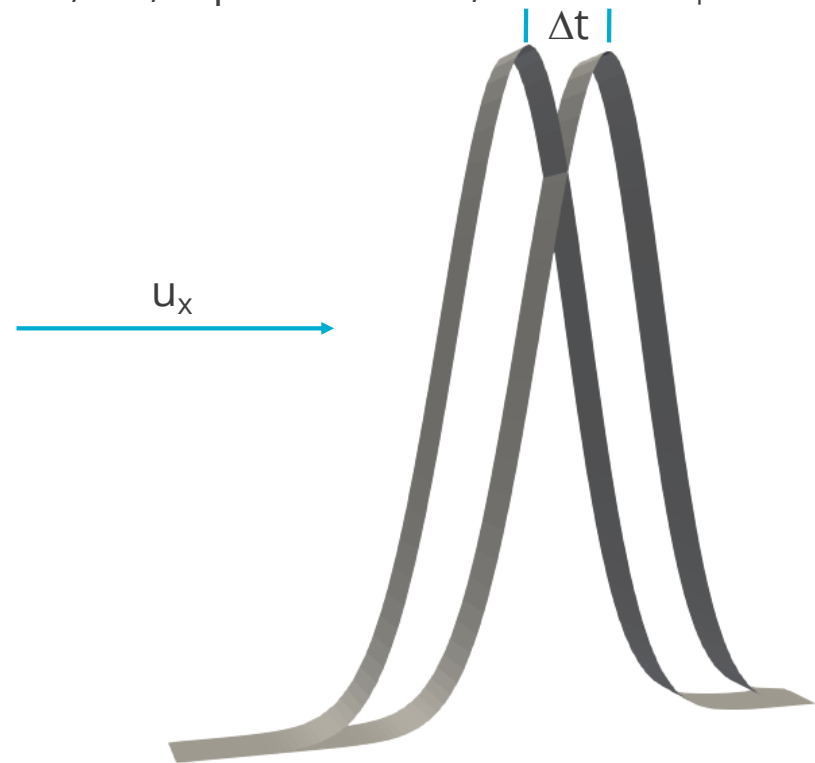
Partial Differential Equation: Transient Evolution of a Scalar with Advection

General Transport Equation using Einstein notation, i.e., repeated index, for scalar ϕ

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- Velocity, u_j [m/s]
- Mass diffusivity, D [m²/s²]
- Dynamic viscosity, μ [kg/(m-s)]
- Schmidt number, Sc

$$\frac{\partial \rho \phi}{\partial t} + \frac{\partial \rho u_j \phi}{\partial x_j} - \frac{\partial}{\partial x_j} \left(\frac{\mu}{Sc} \frac{\partial \phi}{\partial x_j} \right) = S^\phi$$

Two blue arrows point from the terms $\frac{\mu}{Sc} \frac{\partial \phi}{\partial x_j}$ and S^ϕ to the value 0.0.





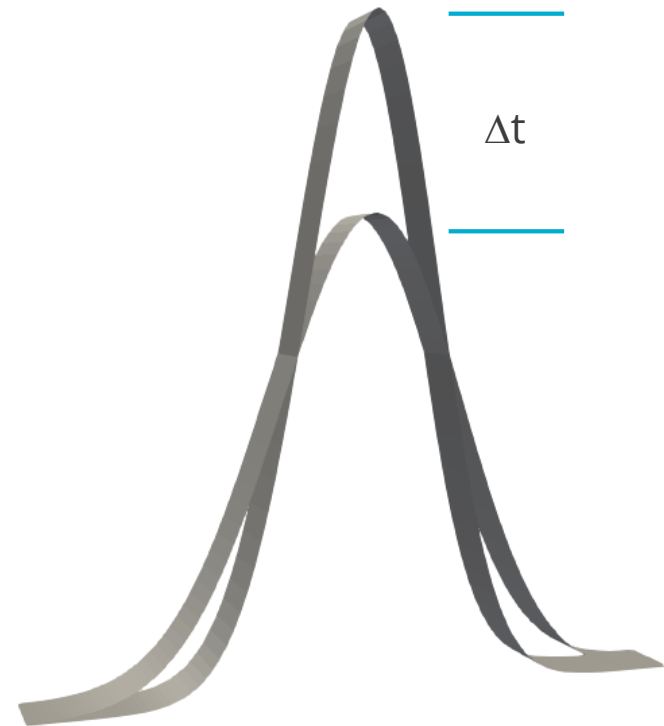
Partial Differential Equation: Transient Evolution of a Scalar with Diffusion

General Transport Equation using Einstein notation, i.e., repeated index, for scalar ϕ

- Density, ρ [kg/m³]
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$$\frac{\partial \rho \phi}{\partial t} + \frac{\partial \rho u_j \phi}{\partial x_j} - \frac{\partial}{\partial x_j} \left(\frac{\mu}{Sc} \frac{\partial \phi}{\partial x_j} \right) = S^\phi$$

Diagram illustrating the transient evolution of a scalar with diffusion. Two overlapping bell-shaped curves represent the scalar distribution at two different times. The vertical distance between the peaks of the two curves is labeled Δt , indicating the time interval. Two blue arrows point from the '0.0' labels on the x-axis to the corresponding terms in the equation above: one arrow points to the $\frac{\partial \rho \phi}{\partial t}$ term, and the other points to the S^ϕ term.





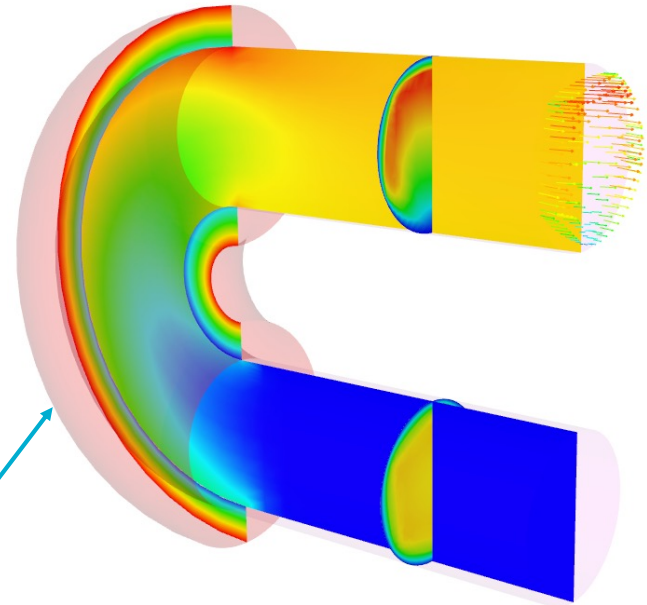
Partial Differential Equation: Transient Heat Conduction

Thermal Heat Conduction

- Density, ρ [kg/m³]
- Specific Heat, C_p [J/(kg-K)]
- Thermal conductivity, λ [W/(m-K)], 1W = 1 J/s
- Heat flux vector, q_j

$$\rho C_p \frac{\partial T}{\partial t} + \frac{\partial q_j}{\partial x_j} = S$$

$$q_j = -\lambda \frac{\partial T}{\partial x_j}$$



Thermal heat conduction (outer shell heated)



Variable Density, non-Isothermal Fluid Flow System

- Consider the variable density (non-isothermal) equations of motion (momentum and continuity) with energy transport:

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho u_j}{\partial x_j} = 0,$$

$$\frac{\partial \rho u_i}{\partial t} + \frac{\partial \rho u_j u_i}{\partial x_j} + \frac{\partial P}{\partial x_i} = \frac{\partial \tau_{ij}}{\partial x_j} + \rho g_i,$$

$$\frac{\partial \rho E}{\partial t} + \frac{\partial \rho u_j H}{\partial x_j} = -\frac{\partial q_j}{\partial x_j} + \frac{\partial u_i \tau_{ij}}{\partial x_j} + \rho u_i g_i$$

$$\rho = \frac{PM}{RT} \quad \text{Equation of State (EOS) provides the P and } \rho \text{ relationship}$$

Constitutive Relationships

$$E = H - P/\rho,$$

$$H = h + \frac{1}{2} u_k u_k,$$

$$\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} \mu \frac{\partial u_k}{\partial x_k} \delta_{ij},$$

$$q_i = -k \frac{\partial T}{\partial x_i},$$

$$h = \int_{T_o}^T C_p dT$$



Conserved vs Non-conserved Form

For the so-called non-conserved form, chain-rule the advection term, while enforcing continuity:

$$\frac{\partial \rho \phi}{\partial t} + \frac{\partial \rho u_j \phi}{\partial x_j} - \frac{\partial}{\partial x_j} \left(\frac{\mu}{Sc} \frac{\partial \phi}{\partial x_j} \right) = S^\phi$$

$$\rho \frac{\partial \phi}{\partial t} + \rho u_j \frac{\partial \phi}{\partial x_j} + \phi \left(\frac{\partial \rho}{\partial t} + \frac{\partial \rho u_j}{\partial x_j} \right) - \frac{\partial}{\partial x_j} \left(\frac{\mu}{Sc} \frac{\partial \phi}{\partial x_j} \right) = S^\phi$$

0.0

$$\longrightarrow \rho \frac{\partial \phi}{\partial t} + \rho u_j \frac{\partial \phi}{\partial x_j} - \frac{\partial}{\partial x_j} \left(\frac{\mu}{Sc} \frac{\partial \phi}{\partial x_j} \right) = S^\phi$$



Incompressible Navier Stokes

For a constant density and viscosity (in absence of a source term), we revert to the classic form:

$$\cancel{\frac{\partial \rho}{\partial t}}^{0.0} + \frac{\partial \rho u_j}{\partial x_j} = 0,$$

$$\frac{\partial \rho u_i}{\partial t} + \frac{\partial \rho u_j u_i}{\partial x_j} + \frac{\partial P}{\partial x_i} = \frac{\partial \tau_{ij}}{\partial x_j}$$

$$\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} \mu \cancel{\frac{\partial u_k}{\partial x_k}}^{0.0} \delta_{ij},$$

$$\frac{\partial u_j}{\partial x_j} = 0,$$

$$\rho \frac{\partial u_i}{\partial t} + \rho \frac{\partial u_j u_i}{\partial x_j} - \mu \frac{\partial^2 u_i}{\partial x_j^2} = - \frac{\partial P}{\partial x_i}$$

$$\text{(trick)} \quad \frac{\partial}{\partial x_j} \mu \frac{\partial u_j}{\partial x_i} = \mu \frac{\partial}{\partial x_i} \cancel{\frac{\partial u_j}{\partial x_j}}^{0.0}$$